

James Durkin

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Education

University of Georgia | Athens, GA

Bachelor of Science in Computer Science | GPA: 3.72 | December 2025

Relevant Coursework: Computer Architecture, Operating Systems, Software Engineering, Data Structures, Networks, Web Programming

Kennesaw State University | Marietta, GA

Master of Science in Electrical and Computer Engineering | December 2027

Technical Projects

STM32 Embedded Systems Library | C, STM32, SPI | Fall 2025

- Engineered a C library for the MCP23S17 IO Expander to simplify and standardize SPI communication with an STM32 microcontroller.

Operating Systems - Thread Synchronization & IPC | C, POSIX, Linux | Fall 2025

- Implemented thread synchronization mechanisms (mutexes, semaphores, condition variables) for multi-threaded embedded applications
- Developed inter-process communication systems using Linux sockets and message queues
- Applied real-time programming principles for concurrent system design

GitHub Copilot Interactive Learning Suite | TypeScript, VS Code API, Git, CI/CD | 2025

- Developed a VS Code extension to teach AI-assisted programming through interactive challenges, featuring persistent progress tracking.
- Architected a professional CI/CD pipeline and demonstrated expert Git workflows and testing strategies for extension development.

Cinema E-Booking System | Java, Spring Boot, MongoDB, JavaScript | Fall 2025

- Engineered a full-stack cinema booking platform with secure authentication/authorization, a payment system with promo codes, and an interactive seat selection UI with real-time availability. Applied an Agile development process.

Autonomous Robot Navigation Simulation | Python, ROS2, Gazebo | Fall 2024

- Implemented SLAM and an Extended Kalman Filter within ROS2 to enable autonomous navigation and real-time state estimation.
- Programmed autonomous navigation for a robot in a simulated Gazebo environment using sensor fusion (LIDAR, IMU).

Experience

ArcGIS Development Team Member | University of Georgia Botanical Garden | Fall 2025

- Developed a public-facing ArcGIS application using Experience Builder and SQL to centralize plant/trail data, featuring a dynamic filtering system with a simplified toggle-based UI for end-users: <https://experience.arcgis.com/experience/f1e5558ce324467fa4ef3fe692b335cd>

Technical Skills

- Languages:** C/C++, Python, Java, JavaScript/TypeScript, C#, SQL, HTML/CSS
- Embedded & Real-Time Systems:** Embedded Systems, Real-time Programming, POSIX/Linux, Multithreading (Mutexes, Semaphores), Kernel-level Synchronization, Microcontroller Development